

# AP-E3 APS Generators and Semisimple Global Forms: Product/Defect Mod-Two Indices, Regulator Non-Uniqueness, and a Simply Connected $Sp(4)$ Candidate

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## Abstract

We evaluate the two reduced generators of  $\Omega_4^{\text{Spin}}(S^3 \times S^2)$  in explicit solvable fermionic models and systematically search semisimple global forms that evade the previous  $SU(3)$  charge obstruction. The generators are

$$G_3 = (S_R^1 \times S^3, \text{pr}_{S^3}, \text{pt}), \quad G_2 = (T_{RR}^2 \times S^2, \text{pt}, \text{pr}_{S^2}).$$

The periodic-circle real Dirac operator has one zero mode and the odd torus chiral Dirac operator has mod-two index one. A codimension-three defect regulator therefore evaluates  $(-1)^{\nu_3}$ , while an Arf-decorated codimension-two regulator evaluates  $(-1)^{\nu_2}$ . Stacking these two invertible regulators produces all four character values on  $(G_3, G_2)$ . By contrast, the untwisted ambient product Dirac spectra are chirality paired and provide only a reference phase  $+1$  in a factorized regulator.

This is a controlled negative result: it computes a product/defect mod-two index character table, not the unique eta phase of the charged two-colour UV theory. The current data do not specify the complete Euclidean Dirac–Yukawa operator, Pauli–Villars mass signs, or five-dimensional mapping-torus APS problem. Heavy invertible spectators can independently flip both torsion signs without changing the degree-five differential curvature. Hence neither sign is fixed from the present effective action.

For global forms, a short theorem shows that any unbroken  $[SU(2)_c \times U(1)]/\mathbb{Z}_2$  excludes a charge-one colour singlet, although it admits  $\mathbf{2}_{+1}$ . Thus the diagonal quotient itself, not merely  $SU(3)$  branching, is the obstruction. Three unquotiented semisimple forms pass the charge screen. The strongest simple candidate in the scanned set is the simply connected group  $Sp(4) \equiv USp(4) \simeq Spin(5)$ , with

$$\mathbf{5} \rightarrow \mathbf{2}_{+1} \oplus \mathbf{2}_{-1} \oplus \mathbf{1}_0, \quad \mathbf{4} \rightarrow \mathbf{2}_0 \oplus \mathbf{1}_{+1} \oplus \mathbf{1}_{-1}.$$

Two Dirac  $\mathbf{5}$ ’s and two complex scalar  $\mathbf{4}$ ’s give the required two-flavour quarks and an exact copy-space  $SU(2)_\phi$  doublet. One real adjoint can break to  $SU(2)_c \times U(1)$  and split all displayed partners at tree level. The one-loop coefficient is  $b_0^{Sp(4)} = 15/2 > 0$ , and the displayed vectorlike spectrum passes local and Witten-parity checks. Nevertheless, the massless charged component is tuned, the heavy eta threshold is unmatched, and no  $Sp(4)$  strong-phase or soliton calculation is supplied. The exactly-two operator is therefore realized only at the group-theory level; the exactly-two gauge/unit-cell rule is not rederived. No Route-E or degree-one-portal promotion is made.

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## 1 Scope and claim boundary

The previous global audit separated the unique ordinary degree-five differential character from two spin-torsion characters. With  $X = S^3 \times S^2$ , stable splitting gives

$$\Omega_4^{\text{Spin}}(X) \simeq \mathbb{Z} \oplus \mathbb{Z}_2 \oplus \mathbb{Z}_2, \quad \tilde{\Omega}_4^{\text{Spin}}(X) \simeq \mathbb{Z}_2 \oplus \mathbb{Z}_2. \quad (1)$$

After the field-independent gravitational factor is removed, the general spin refinement has the form

$$Z_{n,\epsilon_3,\epsilon_2} = Z_n^{\text{diff}}(-1)^{\epsilon_3\nu_3+\epsilon_2\nu_2}, \quad (\epsilon_3, \epsilon_2) \in \mathbb{Z}_2^2. \quad (2)$$

The purpose of this note is to answer two narrower questions:

- (Q1) what can be computed on explicit representatives of the two reduced generators with a fully stated product/defect regulator;
- (Q2) which low-rank semisimple global forms contain both  $q \sim \mathbf{2}_{+1}$  and  $\phi \sim \mathbf{1}_{+1}$ , so that  $qq(\phi^\dagger)^2$  is neutral.

Three distinctions will be maintained throughout.

1. A paired spectrum of an untwisted four-dimensional product Dirac operator is a *reference spectral control*; it is not, by itself, a five-dimensional Dai–Freed computation.
2. A defect mod-two index computes the phase of the explicitly selected invertible regulator. It proves that a torsion character is realizable, not that charged QC2D selects it.
3. Charge neutrality and branching prove an operator-level selection rule. They do not derive the microscopic exactly-two gauge/unit-cell constraint, threshold naturalness, or the nonperturbative phase.

This discipline follows the APS/Dai–Freed treatment of determinant phases [1, 2, 3] and the bordism approach to four-dimensional anomalies [4, 5].

## 2 Explicit representatives of both reduced generators

### 2.1 Pontryagin–Thom invariants

For a sigma-model field  $\Phi = (g, s) : M_4 \rightarrow S^3 \times S^2$ , choose regular values  $p_3 \in S^3$ ,  $p_2 \in S^2$ . Their transverse inverse images are

$$L_g = g^{-1}(p_3), \quad \dim L_g = 1, \quad \Sigma_s = s^{-1}(p_2), \quad \dim \Sigma_s = 2. \quad (3)$$

The target orientations frame the normal bundles. Together with the spin structure on  $M_4$ , they induce spin structures on  $L_g$  and  $\Sigma_s$ . The reduced invariants are

$$\nu_3 = [L_g] \in \Omega_1^{\text{Spin}} \simeq \mathbb{Z}_2, \quad \nu_2 = \text{Arf}(\Sigma_s) \in \Omega_2^{\text{Spin}} \simeq \mathbb{Z}_2. \quad (4)$$

Take unit-radius factors and define

$$G_3 := (S_{\text{R}}^1 \times S^3, g = \text{pr}_{S^3}, s = \text{const}), \quad (5)$$

$$G_2 := (T_{\text{RR}}^2 \times S^2, g = \text{const}, s = \text{pr}_{S^2}). \quad (6)$$

Here R denotes periodic spin structure. The inverse images are

$$L_g = S_{\text{R}}^1 \times \{p_3\}, \quad \Sigma_s = T_{\text{RR}}^2 \times \{p_2\}. \quad (7)$$

The periodic circle is the nonbounding generator of  $\Omega_1^{\text{Spin}}$ , while the RR torus is the odd spin surface with Arf invariant one [6]. Therefore

$$(\nu_3(G_3), \nu_2(G_3)) = (1, 0), \quad (\nu_3(G_2), \nu_2(G_2)) = (0, 1). \quad (8)$$

### 2.2 Circle spectrum and mod-two spectral flow

Let the circle have circumference  $2\pi$ . Its real Dirac spectrum is

$$D_{S^1, \alpha} = -i \frac{d}{dx}, \quad \lambda_n = n + \alpha, \quad n \in \mathbb{Z}, \quad (9)$$

where  $\alpha = 0$  for R and  $\alpha = 1/2$  for NS spin structure. Hence

$$\dim_{\mathbb{R}} \ker D_{S^1, 0} = 1 \pmod{2}, \quad \ker D_{S^1, 1/2} = 0. \quad (10)$$

For the explicit mass path

$$H_{\alpha}(t) = D_{S^1, \alpha} + t, \quad -\frac{1}{4} \leq t \leq \frac{1}{4}, \quad (11)$$

only the  $n = 0$  R eigenvalue crosses zero. Thus

$$\text{SF}_2(H_0) = 1, \quad \text{SF}_2(H_{1/2}) = 0, \quad (12)$$

and APS identifies this with the mod-two index of  $\partial_t + H_{\alpha}(t)$ . A real Pfaffian changes sign exactly in the R case.

### 2.3 Odd-torus chiral index and Arf phase

On a square torus of side  $2\pi$ , the chiral Dirac eigenvalues may be normalized as

$$D_{\alpha\beta}^+ : \quad \lambda_{mn} = (m + \alpha) + i(n + \beta), \quad (m, n) \in \mathbb{Z}^2. \quad (13)$$

Only  $(\alpha, \beta) = (0, 0)$  has a zero mode. Consequently

$$\text{ind}_2 D_{T_{\text{RR}}^2}^+ = \dim_{\mathbb{C}} \ker D_{00}^+ \pmod{2} = 1, \quad \text{ind}_2 D_{\alpha\beta}^+ = 0 \quad \text{otherwise.} \quad (14)$$

The equality

$$\text{Arf}(T^2, \alpha, \beta) = \text{ind}_2 D_{\alpha\beta}^+ \quad (15)$$

identifies the Majorana Pfaffian sign on  $G_2$ . This is a KO/mod-two index statement and needs no continuum extrapolation.

### 3 Reference ambient spectra versus the actual APS problem

#### 3.1 Paired product spectra

The ordinary Dirac spectra on unit spheres are

$$\text{spec}(D_{S^3}) = \{\pm(k + \frac{3}{2}), \quad \text{mult} = (k+1)(k+2)\}_{k \geq 0}, \quad (16)$$

$$\text{spec}(D_{S^2}) = \{\pm(\ell + 1), \quad \text{mult} = 2(\ell + 1)\}_{\ell \geq 0}. \quad (17)$$

For product metrics,

$$|\lambda(D_{S^1_{\mathbb{R}} \times S^3})|^2 = n^2 + (k + \frac{3}{2})^2 \geq \frac{9}{4}, \quad (18)$$

$$|\lambda(D_{T^2_{\mathbb{R}\mathbb{R}} \times S^2})|^2 = m^2 + n^2 + (\ell + 1)^2 \geq 1. \quad (19)$$

Both ambient four-dimensional operators are gapped, and chirality pairs every nonzero eigenvalue with its negative. A symmetric, factorized Pauli–Villars convention therefore assigns reference phase +1 to both products.

*Remark 3.1* (What the paired spectrum does not prove). The eta invariant controlling a four-dimensional microscopic anomaly belongs to a specified odd-dimensional mapping-torus or APS problem. Equations (18)–(19) do not determine that problem. They only verify that an untwisted ambient product determinant has no unpaired four-dimensional eigenvalue. Calling this calculation “the charged-QC2D eta invariant” would be incorrect.

#### 3.2 An explicit defect Dirac–Yukawa family

We now construct a separate, computable regulator family. In a tubular neighbourhood of a transverse inverse image  $N = \Phi^{-1}(p)$ , use oriented normal coordinates  $u^a$ ,  $a = 1, \dots, r$ , and an auxiliary real Clifford module with matrices  $\Gamma_a$ . A local Dirac–Yukawa operator is

$$\mathcal{D}_{r,\mu} = \not{D}_{M_4} \otimes \mathbf{1} + \mu \sum_{a=1}^r u^a(\Phi) \Gamma_a. \quad (20)$$

Using a cutoff in the tubular neighbourhood and a positive Pauli–Villars mass outside gives a global Fredholm family. The normal harmonic-oscillator modes pair except for a single localized mode on  $N$ , as in the Jackiw–Rebbi mechanism [7]. Gapping that localized mode by the nontrivial one- or two-dimensional invertible spin phase gives

$$R_3[M_4, g] = (-1)^{\text{ind}_2 D_{L_g}} = (-1)^{\nu_3(M_4, g)}, \quad (21)$$

$$R_2[M_4, s] = (-1)^{\text{ind}_2 D_{\Sigma_s}^+} = (-1)^{\text{Arf}(\Sigma_s)} = (-1)^{\nu_2(M_4, s)}. \quad (22)$$

The full APS operator factorizes in the large- $\mu$  limit into paired normal modes and the defect operator. Therefore its mod-two index is the right-hand side of Eqs. (21)–(22). This establishes an explicit regulator realization of each character.

The resulting character table is

| $(\epsilon_3, \epsilon_2)$ | regulator stack | phase on $G_3$ | phase on $G_2$ |
|----------------------------|-----------------|----------------|----------------|
| (0, 0)                     | reference       | +1             | +1             |
| (1, 0)                     | $R_3$           | −1             | +1             |
| (0, 1)                     | $R_2$           | +1             | −1             |
| (1, 1)                     | $R_3 R_2$       | −1             | −1             |

**Theorem 3.2** (Regulator non-uniqueness). *The differential curvature and all perturbative local terms held fixed do not determine  $(\epsilon_3, \epsilon_2)$ . Stacking either heavy invertible defect regulator changes exactly one torsion sign. Hence a microscopic UV Dirac–Yukawa operator and its regulator are necessary data for selecting the two signs.*

*Proof.* Equations (21) and (22) are homomorphisms from reduced spin bordism to  $\{\pm 1\}$ . Equation (8) shows that their evaluation matrix is the identity over  $\mathbb{Z}_2$ , so they form a basis of  $\text{Hom}(\mathbb{Z}_2 \oplus \mathbb{Z}_2, U(1))$ . They are torsion phases and have zero de Rham curvature. Stacking therefore leaves the fixed degree-five curvature unchanged while toggling either sign.  $\square$

**Fail-closed gate 3.3** (Charged-QC2D determinant). The product/defect construction proves the four possible phase assignments and supplies a regulator-dependence negative control. The current Route-F files do not specify the complete charged-QC2D Euclidean operator

$$\mathcal{D}_{UV} = \gamma^\mu (\nabla_\mu + A_\mu) + P_L \mathcal{M}(g, s) + P_R \mathcal{M}(g, s)^\dagger, \quad (23)$$

including every heavy representation, Yukawa matrix, Pauli–Villars mass sign, and mapping-torus extension. Thus

$$\begin{aligned} \text{epsilon\_3\_selected\_by\_charged\_qc2d\_uv} &= \text{false}, \\ \text{epsilon\_2\_selected\_by\_charged\_qc2d\_uv} &= \text{false}. \end{aligned} \quad (24)$$

## 4 A global-form theorem before branching

Let

$$H_0 = SU(2)_c \times U(1)_H, \quad H_1 = \frac{SU(2)_c \times U(1)_H}{\langle (-\mathbf{1}, e^{i\pi}) \rangle}. \quad (25)$$

Normalize  $U(1)_H$  so that its smallest desired charge is one. A term  $(j, q)$  descends to  $H_1$  precisely when the quotient generator acts trivially:

$$(-1)^{2j}(-1)^q = 1 \iff 2j + q = 0 \pmod{2}. \quad (26)$$

**Theorem 4.1** (Diagonal-quotient obstruction). *The diagonal quotient  $H_1$  admits a colour doublet  $\mathbf{2}_{+1}$  but excludes a colour singlet  $\mathbf{1}_{+1}$ . Therefore any semisimple embedding with unbroken global form  $H_1$  fails the charge-one/exactly-two field requirement independently of its Lie algebra branching.*

*Proof.* For  $\mathbf{2}_{+1}$ ,  $2j + q = 1 + 1 = 2$ , so Eq. (26) holds. For  $\mathbf{1}_{+1}$ , it gives  $0 + 1 = 1$ , so it fails. Since this is a statement about the kernel of the group homomorphism, no higher representation can evade it.  $\square$

The exact operator charge condition is

$$Q \left[ qq(\phi^\dagger)^m \right] = 2X_q - mX_\phi. \quad (27)$$

With  $X_q = X_\phi = 1$ , the first neutral power is  $m = 2$ . Thus an *unquotiented*  $H_0$  solves the operator charge equation. It does not by itself prove that a microscopic unit cell has exactly two constituents.

## 5 Systematic low-rank semisimple search

### 5.1 Baseline failures

For  $SU(3) \rightarrow [SU(2)_c \times U(1)]/\mathbb{Z}_2$ ,

$$\mathbf{3} \rightarrow \mathbf{2}_{+1} \oplus \mathbf{1}_{-2}. \quad (28)$$

The quotient theorem proves that no  $\mathbf{1}_{+1}$  occurs in any  $SU(3)$  representation. This recovers the previous no-go without relying on a finite representation scan.

The same warning applies to covers and quotients. The direct product  $SU(2)_c \times SU(2)_H$  contains the required representations, whereas  $SO(4) = [SU(2)_c \times SU(2)_H]/\mathbb{Z}_2$  removes the scalar. Likewise the spinor  $\mathbf{4}$  of  $Spin(5)$  does not descend to  $SO(5)$ . Global form is therefore physical data, not notation.

## 5.2 Minimal product candidate

Take the unquotiented semisimple group

$$G_{\text{prod}} = SU(2)_c \times SU(2)_H \xrightarrow{\langle \Sigma_H \rangle} SU(2)_c \times U(1)_H, \quad (29)$$

where a real  $(\mathbf{1}, \mathbf{3})$  scalar has  $\langle \Sigma_H \rangle = VT_H^3$ , and the doublet weights are  $q = \pm 1$ . Then

$$(\mathbf{2}, \mathbf{2}) \rightarrow \mathbf{2}_{+1} \oplus \mathbf{2}_{-1}, \quad (30)$$

$$(\mathbf{1}, \mathbf{2}) \rightarrow \mathbf{1}_{+1} \oplus \mathbf{1}_{-1}. \quad (31)$$

Two Dirac bifundamentals supply two low-energy quark flavours. Two complex scalar  $(\mathbf{1}, \mathbf{2})$  copies  $\Phi_I$ ,  $I = 1, 2$ , retain an exact copy-space  $SU(2)_\phi$  when their couplings are proportional to  $\delta_{IJ}$ . The two desired charge-one components then form the  $SU(2)_\phi$  doublet used in  $\text{Sym}^2 \mathbf{2} = \mathbf{3}$ .

The group is already a viable minimal semisimple backup. Its limitation is that each required field comes with an opposite-charge partner, and the massless branch requires the threshold split discussed below.

## 5.3 Best simple candidate in the scanned set: simply connected $Sp(4)$

We use physics notation

$$Sp(4) \equiv USp(4) \simeq Spin(5), \quad (32)$$

whose fundamental has dimension four. It contains  $Spin(4) = SU(2)_c \times SU(2)_H$ . Under this subgroup

$$\mathbf{4} \rightarrow (\mathbf{2}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2}), \quad (33)$$

$$\mathbf{5} \rightarrow (\mathbf{2}, \mathbf{2}) \oplus (\mathbf{1}, \mathbf{1}), \quad (34)$$

$$\mathbf{10} \rightarrow (\mathbf{3}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{3}) \oplus (\mathbf{2}, \mathbf{2}). \quad (35)$$

These formulas also follow directly from  $\mathbf{5} = \wedge^2 \mathbf{4} - \mathbf{1}$  and  $\mathbf{10} = \text{Sym}^2 \mathbf{4}$  [8]. Breaking the second factor by a real adjoint  $\Sigma \in \mathbf{10}$ , with its vacuum in the  $(\mathbf{1}, \mathbf{3})_0$  direction, gives

$$\mathbf{4} \rightarrow \mathbf{2}_0 \oplus \mathbf{1}_{+1} \oplus \mathbf{1}_{-1}, \quad (36)$$

$$\mathbf{5} \rightarrow \mathbf{2}_{+1} \oplus \mathbf{2}_{-1} \oplus \mathbf{1}_0, \quad (37)$$

$$\mathbf{10} \rightarrow \mathbf{1}_{+2} \oplus \mathbf{2}_{+1} \oplus \mathbf{3}_0 \oplus \mathbf{1}_0 \oplus \mathbf{2}_{-1} \oplus \mathbf{1}_{-2}. \quad (38)$$

The subgroup embedding is injective. In particular,  $(-\mathbf{1}_c, -\mathbf{1}_H)$  maps to the nontrivial central element of  $Spin(5)$ , not to the identity. After  $SU(2)_H \rightarrow U(1)_H$ , the unbroken global form is therefore  $H_0$ , so both odd-charge representations are legitimate. If one instead quotients  $Spin(5)$  to  $SO(5)$ , this central element is identified with the identity and the spinor  $\mathbf{4}$  is lost. The scalar requirement forces the simply connected cover.

An explicit field list is

$$\Psi_i \in \mathbf{5} \text{ Dirac}, \quad i = 1, 2; \quad \Phi_I \in \mathbf{4} \text{ complex}, \quad I = 1, 2; \quad \Sigma \in \mathbf{10} \text{ real}. \quad (39)$$

The two  $\Phi_I$  copies are essential. Their copy index carries the exact global  $SU(2)_\phi$ ; the two gauge weights inside a single  $\mathbf{4}$  are not a substitute for this flavour doublet. With copy-symmetric couplings, the light  $\mathbf{1}_{+1}$  components are  $\phi_I$ , and

$$qq(\phi_I^\dagger \phi_J^\dagger)_{\text{sym}} \in \text{Sym}^2 \mathbf{2}_\phi = \mathbf{3}_\phi. \quad (40)$$

Equation (40) establishes the desired group representation of the local operator. It does not prove the exactly-two microscopic Gauss rule.

## 5.4 A secondary simple candidate: $SU(4)$

The block embedding

$$\iota_4(U, z) = \text{diag}(zU, z^{-1}, z^{-1}) \in SU(4) \quad (41)$$

has trivial kernel. Consequently

$$\mathbf{4} \rightarrow \mathbf{2}_{+1} \oplus 2\mathbf{1}_{-1}, \quad (42)$$

$$\bar{\mathbf{4}} \rightarrow \mathbf{2}_{-1} \oplus 2\mathbf{1}_{+1}. \quad (43)$$

Two suitably oriented adjoint vacua can reduce the centralizer to the direct  $SU(2)_c \times U(1)$  while preserving the block charges. The fundamental also has a useful uniqueness property. For an  $SU(N)$  block embedding with doublet charge  $a$  and each of the other  $N - 2$  entries of charge  $b$ , tracelessness gives

$$2a + (N - 2)b = 0. \quad (44)$$

Demanding equal-magnitude quark and conjugate-singlet charges,  $a = -b \neq 0$ , forces

$$(4 - N)a = 0 \implies N = 4. \quad (45)$$

Thus  $SU(4)$  is the unique member of this fundamental block family with the desired equality. Quotienting by  $\{-\mathbf{1}_4\}$  forbids both  $\mathbf{4}$  and  $\bar{\mathbf{4}}$ , so again the simply connected form is required. We rank this below  $Sp(4)$  because the required breaking and copy-flavour protection use more independent ingredients.

## 6 Tree-level partner decoupling

### 6.1 Universal adjoint split

In all three passing candidates, the desired and conjugate components have weights  $q = \pm 1$  under the broken generator. A vectorlike Dirac multiplet admits

$$\mathcal{L}_\Psi = -\bar{\Psi}(m + y\Sigma)\Psi, \quad M_q = m + qyV. \quad (46)$$

Choosing

$$m = -yV \quad (47)$$

gives

$$M_+ = 0, \quad M_- = -2yV. \quad (48)$$

For  $Sp(4)$ , the neutral singlet in  $\mathbf{5}$  has mass  $M_0 = -yV$ . Thus all unwanted fermions are heavy at tree level when  $yV \neq 0$ .

For each scalar copy, take

$$V_2 = \Phi_I^\dagger (m_\Phi^2 + \kappa\Sigma) \Phi_I. \quad (49)$$

The benchmark

$$m_\Phi^2 = 1, \quad \kappa V = -2 \quad (50)$$

gives

$$m_{\phi,+}^2 = -1, \quad m_{\phi,-}^2 = 3, \quad m_{\phi,0}^2 = 1 \quad (Sp(4)). \quad (51)$$

Quartics can stabilize the desired charged vacuum while leaving the displayed partners positive at tree level.

**Fail-closed gate 6.1** (Radiative protection). Equation (47) is a cancellation between independent operators. No symmetry protecting it has been specified. Heavy gauge, adjoint, and partner loops generically induce

$$\delta M_+ \sim \frac{g_G^2}{16\pi^2} yV \quad (52)$$

up to representation- and scheme-dependent coefficients. The candidate is therefore an explicit tree-level spectrum, not a natural all-scale decoupling theorem.

## 6.2 Orientation and the unmatched heavy determinant

The retained colour doublet has  $X_q = +1$  and two colours. Its light-field mixed coefficient is therefore

$$k_{\text{light}} = N_c X_q = 2(+1) = +2. \quad (53)$$

This fixes the algebraic orientation convention. Integrating out the  $q = -1$  partner and neutral fields also generates regulator-dependent local and torsion counterterms. Until their determinant phases are matched across the breaking threshold, Eq. (53) is not an all-scale Dai–Freed derivation of  $k = +2$ .

## 7 Gauge anomalies and one-loop running

### 7.1 Displayed anomaly checks

All fermions in the candidate lists are Dirac fields in complete deep-group representations. Their left- and right-handed perturbative gauge anomalies cancel. For the product group, two Dirac  $(\mathbf{2}, \mathbf{2})$ ’s give

$$N_{\text{Weyl doublet}}^{(c)} = N_{\text{Weyl doublet}}^{(H)} = N_f (2 \text{ chiralities}) (2 \text{ spectator dimension}) = 8, \quad (54)$$

which is even. In the  $Sp(4)$  candidate, the displayed Dirac fields double every chiral mod-two phase. After breaking, the two light Dirac colour doublets give four left-Weyl colour doublets, also even. These are the ordinary Witten-parity checks [9, 10].

**Fail-closed gate 7.1** (Full gauge-bordism phase). Vectorlike cancellation and the  $\pi_4$  parity test are necessary checks, not a substitute for evaluating the complete  $\Omega_5^{\text{Spin}}(BSp(4))$  Dai–Freed character with every Higgs and background bundle retained. We therefore set

$$\text{best\_candidate\_full\_dai\_freed\_gauge\_bordism\_audit\_complete=false}. \quad (55)$$

This is deliberately conservative even though the displayed matter is vectorlike.

### 7.2 Asymptotic-freedom screen

For Dirac fermions, real scalars, and complex scalars, respectively, use

$$b_0 = \frac{11}{3}C_2(G) - \frac{4}{3} \sum_{\text{Dirac}} T(R_f) - \frac{1}{6} \sum_{\text{real}} T(R_s) - \frac{1}{3} \sum_{\text{complex}} T(R_s). \quad (56)$$

Our conventions are

$$T_{SU(2)}(\mathbf{2}) = \frac{1}{2}, \quad T_{SU(2)}(\mathbf{3}) = 2, \quad T_{Sp(4)}(\mathbf{4}) = \frac{1}{2}, \quad T_{Sp(4)}(\mathbf{5}) = 1, \quad T_{Sp(4)}(\mathbf{10}) = 3. \quad (57)$$

For the product candidate, the bifundamental spectator dimension makes  $\sum_f T = 2$  for each factor. With one real  $SU(2)_H$  adjoint and two complex  $H$  doublets,

$$b_{0,c}^{\text{prod}} = \frac{22}{3} - \frac{4}{3}(2) = \frac{14}{3} > 0, \quad (58)$$

$$b_{0,H}^{\text{prod}} = \frac{22}{3} - \frac{4}{3}(2) - \frac{1}{6}(2) - \frac{1}{3} \left( 2 \cdot \frac{1}{2} \right) = 4 > 0. \quad (59)$$

For two Dirac  $\mathbf{5}$ ’s, one real  $\mathbf{10}$ , and two complex  $\mathbf{4}$ ’s of  $Sp(4)$ ,

$$b_0^{Sp(4)} = 11 - \frac{4}{3}(2) - \frac{1}{6}(3) - \frac{1}{3} \left( 2 \cdot \frac{1}{2} \right) = \frac{15}{2} > 0. \quad (60)$$



For the illustrative  $SU(4)$  list of two Dirac fundamentals, two real adjoints needed for the breaking, and two complex antifundamentals,

$$b_0^{SU(4)} = \frac{44}{3} - \frac{4}{3}(1) - \frac{1}{6}(8) - \frac{1}{3}(1) = \frac{35}{3} > 0. \quad (61)$$

All three pass the one-loop asymptotic-freedom screen. This says nothing by itself about which Higgs vacuum is nonperturbatively selected.

## 8 Classification and exact status

| Deep group               | relevant global form | charge screen | conclusion                                                 |
|--------------------------|----------------------|---------------|------------------------------------------------------------|
| $SU(3)$                  | unbroken $H_1$       | fail          | charge-one singlet forbidden for every irrep               |
| $SU(2)_c \times SU(2)_H$ | direct product       | pass          | minimal semisimple backup; opposite-charge partners        |
| $SO(4)$                  | diagonal quotient    | fail          | $(\mathbf{1}, \mathbf{2})$ scalar does not descend         |
| $Sp(4) = Spin(5)$        | simply connected     | pass          | best simple candidate in scanned set                       |
| $SO(5)$                  | $Sp(4)/\mathbb{Z}_2$ | fail          | required spinor $\mathbf{4}$ scalar absent                 |
| $SU(4)$                  | simply connected     | pass          | unique fundamental block equality; more elaborate breaking |
| $SU(4)/\mathbb{Z}_2$     | central quotient     | fail          | $\mathbf{4}, \bar{\mathbf{4}}$ do not descend              |

The word “best” refers only to the scanned algebraic criteria: a single simple simply connected group, injective unbroken subgroup, minimal representations containing both charges, and positive one-loop  $b_0$ . It is not a claim that  $Sp(4)$  dynamically realizes the desired phase.

| Claim                                     | Status                      | Boundary                                                       |
|-------------------------------------------|-----------------------------|----------------------------------------------------------------|
| Product representatives $G_3, G_2$        | proved                      | transverse inverse images generate both reduced summands       |
| Circle and torus mod-two indices          | proved                      | exact Fourier spectra                                          |
| Reference ambient product phase +1        | proved for stated regulator | paired four-dimensional spectrum only                          |
| Four defect-regulator character values    | proved                      | Pontryagin–Thom localization and KO indices                    |
| Unique charged-QC2D eta phase             | open                        | microscopic operator and five-dimensional APS data unspecified |
| Diagonal-quotient obstruction             | proved                      | $2j + q = 0 \pmod{2}$                                          |
| Simply connected $Sp(4)$ charge embedding | proved                      | explicit $\mathbf{4}, \mathbf{5}, \mathbf{10}$ branching       |
| Exactly-two neutral triplet operator      | proved at group level       | needs two complex $\mathbf{4}$ copies; no unit-cell derivation |
| Displayed partner split                   | tree-level                  | $m = -yV$ is unprotected                                       |
| Displayed anomaly and running screen      | passed                      | vectorlike/Witten parity and one-loop $b_0$ only               |
| Full $Sp(4)$ gauge-bordism phase          | open                        | complete Dai–Freed character not evaluated                     |
| All-scale $k = +2$ orientation            | open                        | heavy threshold eta counterterm unmatched                      |

| Claim                                     | Status         | Boundary                                 |
|-------------------------------------------|----------------|------------------------------------------|
| Nonperturbative phase and $B = 1$ soliton | <b>open</b>    | requires new strong-dynamics calculation |
| Degree-one Route-E portal                 | <b>blocked</b> | waits for all preceding gates            |

## 9 Deterministic regression

The companion verifier

`route_f/code/verify_ap_e3_aps_global_form_search.py`

performs the following checks without fitted parameters:

1. counts R/NS circle zero modes and mass-path spectral crossings;
2. counts all four flat-torus spin-structure kernels and verifies the Arf parity;
3. computes the minimum ambient product gaps and a truncated eta-pairing residual;
4. constructs all four defect-regulator character values;
5. enforces the diagonal-centre descent condition;
6. derives the  $Sp(4)$  **5** from  $\wedge^2 \mathbf{4} - \mathbf{1}$  and the **10** from  $\text{Sym}^2 \mathbf{4}$ ;
7. checks the  $Spin(5)$  centre image,  $SU(4)$  block kernel, and the fundamental  $SU(N)$  uniqueness equation;
8. evaluates the displayed mass splits, Witten parities, light level, and one-loop beta coefficients;
9. asserts every fail-closed APS, threshold, dynamics, and portal flag.

The numerical calculations encode and regress the displayed analytic arguments; they do not independently prove a bordism theorem or strong dynamics.

## 10 Ordered decisive calculations

Several strategies remain available, in descending priority.

1. **Microscopic eta route.** Write the full charged-QC2D  $\mathcal{D}_{UV}$  of Eq. (23), including all heavy fields and regulator masses. Construct mapping tori representing  $G_3, G_2$ , then compute the two exponentiated reduced eta invariants. This is the only route that selects  $(\epsilon_3, \epsilon_2)$  for that UV theory.
2. **Simply connected  $Sp(4)$  route.** Derive a symmetry or controlled threshold prescription protecting  $M_+ = 0$ , evaluate the full gauge-bordism character, and match the heavy determinant across  $Sp(4) \rightarrow SU(2)_c \times U(1)$ .
3. **Minimal-product backup.** If the simple-group threshold is obstructed, repeat the determinant calculation in direct  $SU(2)_c \times SU(2)_H$ , retaining the global form and both scalar copies explicitly.
4. **Nonperturbative gate.** For any surviving candidate, establish the charged two-colour vacuum, monopole effects, and the complete  $B = 1$  Hessian before composing with AP-E4.

Until these steps close,

`route_e_uv_completion_closed=false,  
physics_promotion_allowed=false,  
degree_one_portal_allowed=false.`

(62)

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